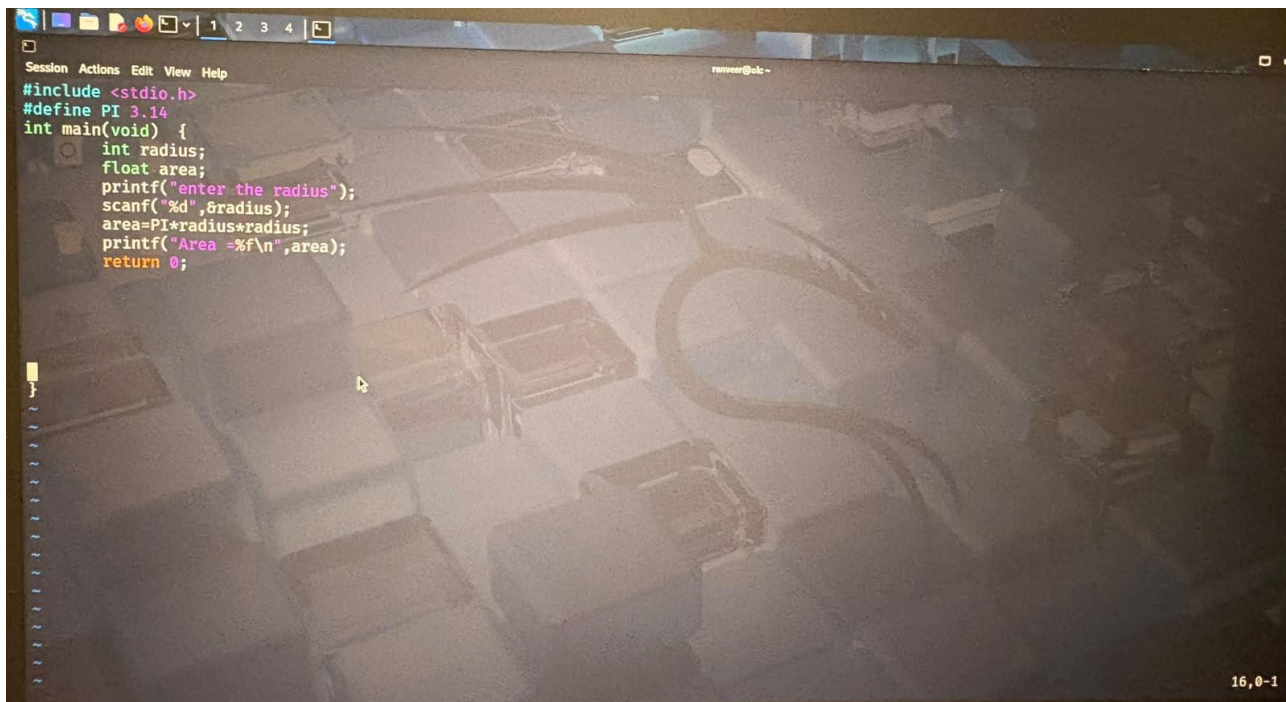


Experiment-2

Demonstration of Stages of Compilation

- **Preprocessing Stage:** Preprocessing is the **first stage** of compilation. The preprocessor handles commands beginning with #, such as #include and #define. It expands header files, replaces macros, and removes comments from the source code.
For example, #include <stdio.h> is processed and the required contents of the header file are included.
Input: area.c
Output: area.i

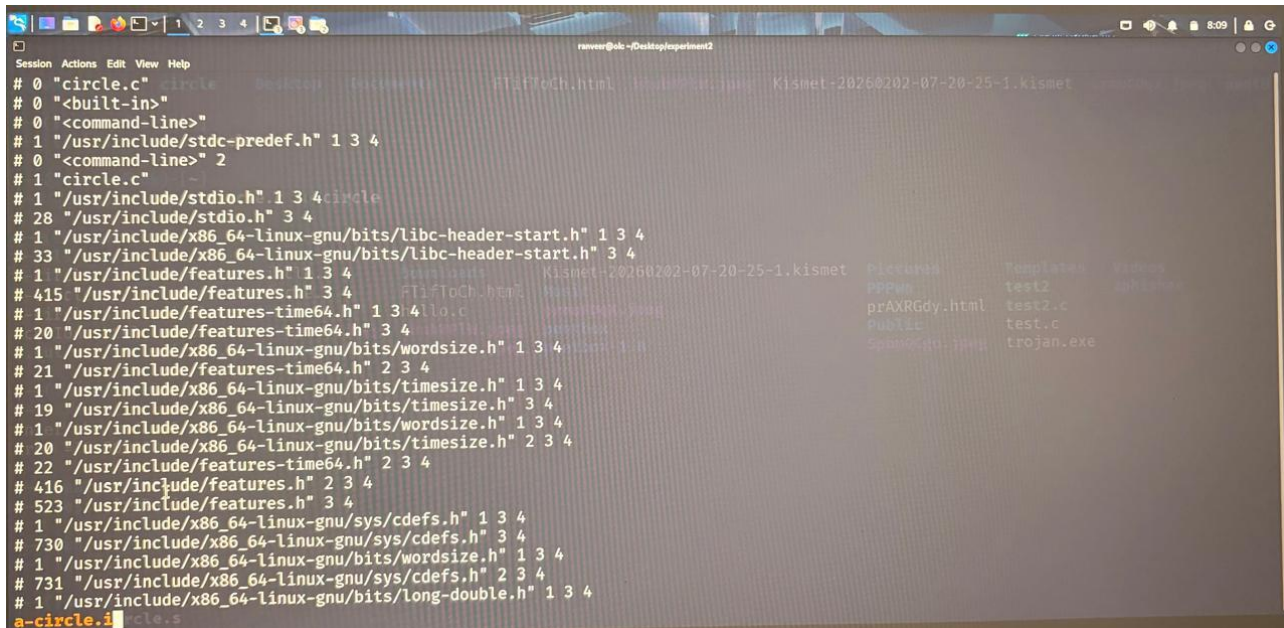
A screenshot of a code editor window with a dark background and light-colored text. The code is written in C and is for calculating the area of a circle. The code includes a preprocessor directive for stdio.h, a macro definition for PI, and a main function that prompts the user for a radius, reads it, calculates the area, and prints it. The code is as follows:

```
Session Actions Edit View Help
#include <stdio.h>
#define PI 3.14
int main(void) {
    int radius;
    float area;
    printf("enter the radius");
    scanf("%d",&radius);
    area=PI*radius*radius;
    printf("Area =%f\n",area);
    return 0;
}
```

The editor has a menu bar at the top with 'Session', 'Actions', 'Edit', 'View', and 'Help'. The status bar at the bottom right shows '16,0-1'.

- **Compilation Stage:** In this stage, the compiler converts the pre-processed C code into assembly language. It checks the program for syntax and other compilation-related errors.
The generated assembly code contains low-level instructions that are closer to the machine language understood by the processor.

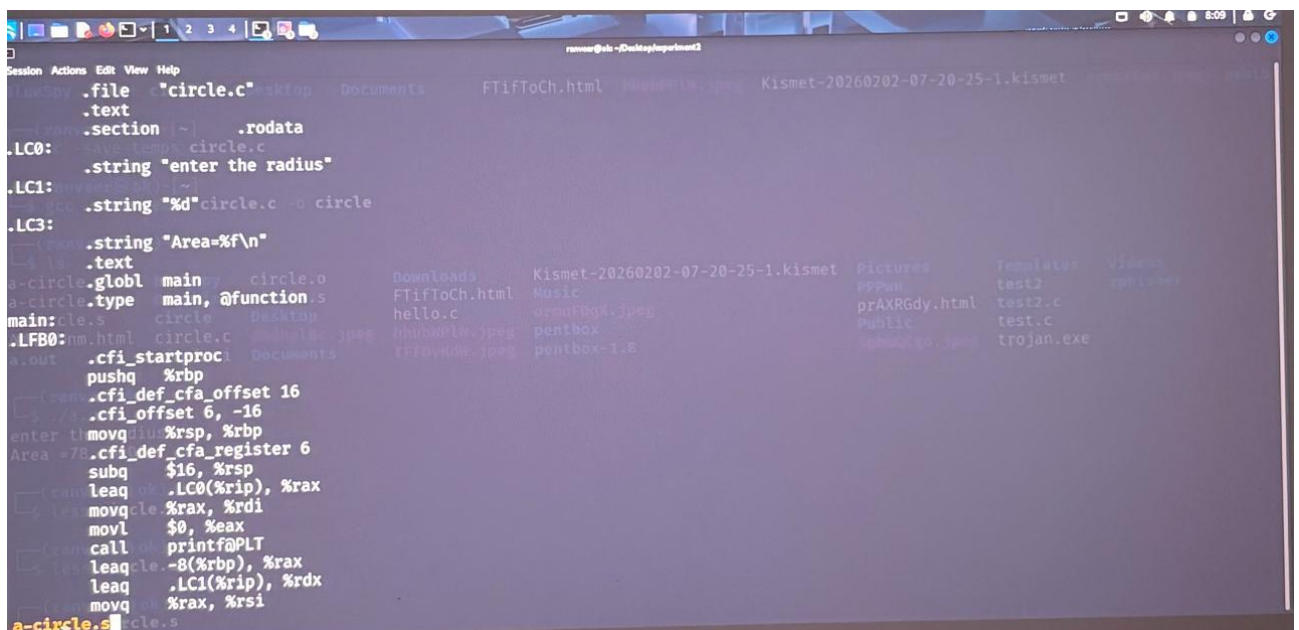
Input: circle.i
Output: circle.s



```
Session Actions Edit View Help
# 0 "circle.c"
# 0 "<built-in>"
# 0 "<command-line>"
# 1 "/usr/include/stdc-predef.h" 1 3 4
# 0 "<command-line>" 2
# 1 "circle.c"
# 1 "/usr/include/stdio.h" 1 3 4
# 28 "/usr/include/stdio.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 1 3 4
# 33 "/usr/include/x86_64-linux-gnu/bits/libc-header-start.h" 3 4
# 1 "/usr/include/features.h" 1 3 4
# 415 "/usr/include/features.h" 3 4
# 1 "/usr/include/features-time64.h" 1 3 4
# 20 "/usr/include/features-time64.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 21 "/usr/include/features-time64.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 1 3 4
# 19 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 20 "/usr/include/x86_64-linux-gnu/bits/timesize.h" 2 3 4
# 22 "/usr/include/features-time64.h" 2 3 4
# 416 "/usr/include/features.h" 2 3 4
# 523 "/usr/include/features.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 1 3 4
# 730 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/wordsize.h" 1 3 4
# 731 "/usr/include/x86_64-linux-gnu/sys/cdefs.h" 2 3 4
# 1 "/usr/include/x86_64-linux-gnu/bits/long-double.h" 1 3 4
a-circle.i circle.s
```

- **Assembly Stage:** The **assembler** converts the assembly language code into **machine-level object code**. This code is stored in an object file. The object file contains machine instructions but is not yet a complete executable program because some functions and libraries may still need to be connected.

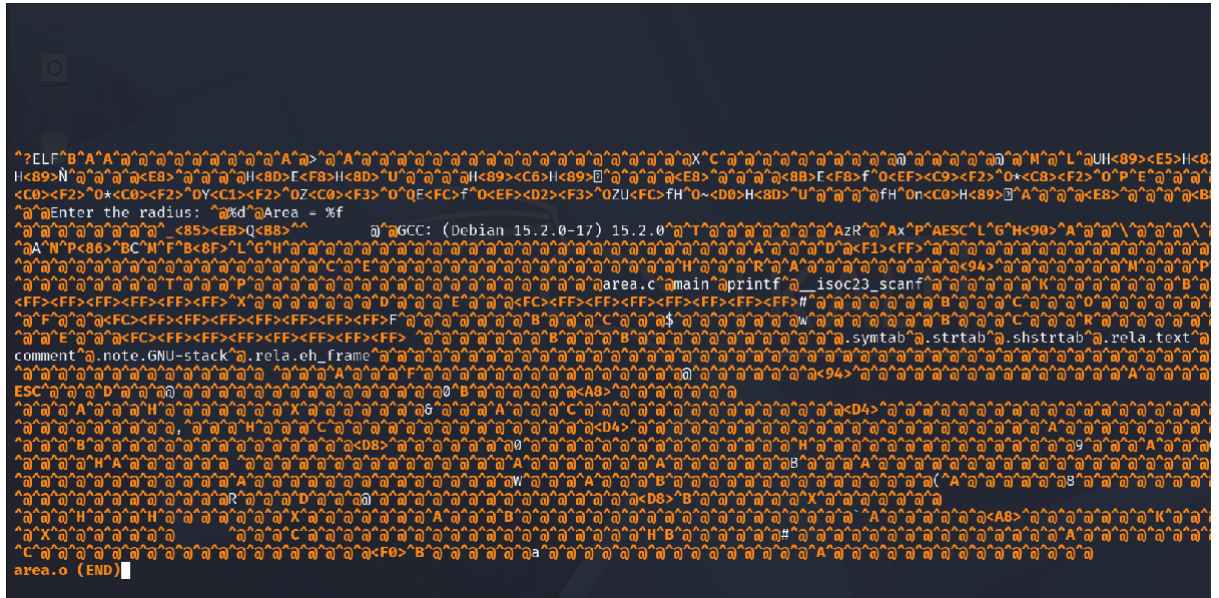
Input: circle.s
Output: circle.o



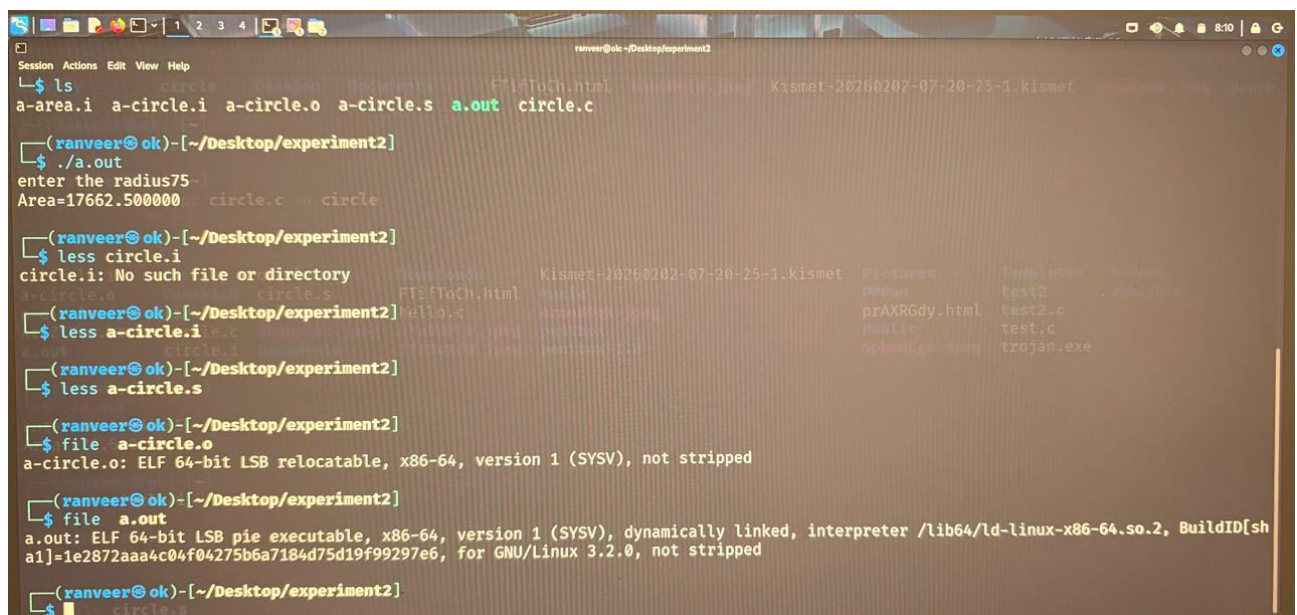
```
Session Actions Edit View Help
.file "circle.c"
.text
.section .rodata
.LC0:
.string "enter the radius"
.LC1:
.string "%d"
.LC3:
.string "Area=%f\n"
.globl main
.type main, @function
main:
.LFB0:
.cfi_startproc
pushq %rbp
.cfi_def_cfa_offset 16
.cfi_offset 6, -16
enter 16,0,%rsp,%rbp
Area = 78
.cfi_def_cfa_register 6
subq $16, %rsp
leaq .LC0(%rip), %rax
movqle %rax, %rdi
movl $0, %eax
call printf@PLT
leaqle -8(%rbp), %rax
leaq .LC1(%rip), %rdx
movqle %rax, %rsi
a-circle.s circle.s
```

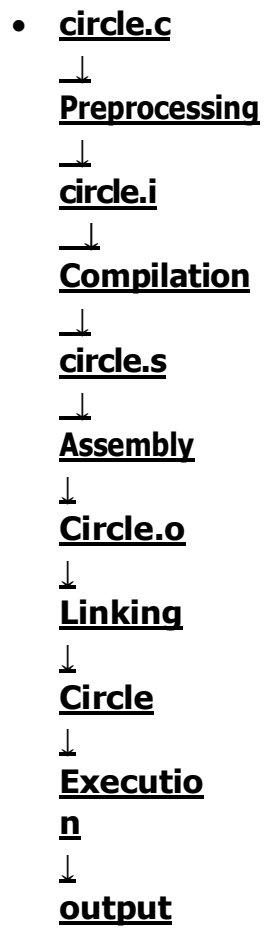

- After linking is completed, a final executable file is created.

Output: circle



- For the circle-area program, the user enters the radius, and the program calculates and displays the area. Radius taken is 5





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